

James Hirst

Location
Email
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[James Hirst](#)

Founding Engineer:

Founding engineer at Mozaic Earth, building a platform that turns mobile-collected biodiversity data into ecologist-verified, AI-enhanced insight. A full-stack and machine learning engineer with Python, TypeScript and Rust expertise, comfortable owning systems end to end and passionate about applying technology to conservation.

Skills:

- | | |
|----------------|--------------|
| • Python | • Azure |
| • Rust | • Kubernetes |
| • React | • Terraform |
| • TypeScript | • TensorFlow |
| • Dart/Flutter | • Supabase |

Experience:

- Founding Engineer, Mozaic Earth (2024-Present)
 - Architecting and managing the Supabase backend - database functions, triggers and edge functions.
 - Leading design and development of the web platform, scoping and reviewing a small team's work.
 - Owning all AI initiatives - building models, data pipelines and researching new technologies.
 - Shipping full-stack features across Supabase, React, Flutter, Figma and Retool.
 - Working with clients on delivery and insight-rich reports via custom SQL workflows.
- Software Engineer, Microsoft (2022-2024)
 - Built a Rust configuration-translation system with an automated testing framework, and a reliable Azure and Kubernetes voicemail solution for a large telco. GitHub Copilot Champion and scrum master for a team of ten.
- Volunteer Machine Learning Engineer, [Shark Trust](#) (2023-Present) classifying shark and ray eggcases by species with convolutional neural networks.
- Machine Learning Engineer, University of Cambridge (2022), applied Detectron2 to delineate tree crowns from drone imagery, improving estimates of carbon stored in forests. Results were published in a peer-reviewed paper.

Qualifications:

- MMath, Mathematics Master's Degree, University of Cambridge, Jesus College (2021-22)
 - Pass with Distinction (79%)
- BA, Mathematics Degree, University of Cambridge, Jesus College (2018-2021)
 - 1st Class in third year (position: 14/222, percentage: 86%)
 - Jesus College Mathematics Award in second year (2019-20) for top performer
- A-Level: 5x A*- Maths, Further Maths, Additional Further Maths, Physics, Chemistry
- IGCSE/ GCSE: 14x A*'s and an A

Other Experience

- Restaurant manager, Hotelplan, winter ski season (2017-18), managed 20 staff of varying ages. Achieved 'Mountain Spirit Award' out of 500 employees for performance.
- Private tutor and Explore Learning Colchester Tutor (2017) in Maths.
- Team Leader for [BL4S](#) (2016), led the winning team for an international competition held by CERN. We proposed an experiment to investigate Einstein's Special Theory of Relativity and then completed the data collection using a particle accelerator at CERN and published a scientific paper in IOP Journal.

Personal Qualifications:

Vice School Captain
BPO Round 1- Gold Award (Top 50 Nationally)
PADI Open Water Certified

Cambridge University Ski Freestyle Captain
Wado-Ryu Karate- 2nd Dan Black belt
National Citizenship Service Award